Minkyun Seo

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Education

Seoul National University (SNU) Sep. 2025 – Present
Integrated Ph.D. Program in Computer Science and Engineering Seoul, Republic of Korea

- Advised by Prof. Jaesik Park at the Visual & Geometric Intelligence Lab (GPA: 4.15/4.3)

Seoul National University (SNU) Mar. 2019 – Aug. 2025
B.S. in Architectural Engineering, Computer Science and Engineering Seoul, Republic of Korea

- **Summa Cum Laude**, GPA: 3.93/4.3 (cumulative), 4.01/4.3 (CSE major)
- **Best BS Thesis Award**: Towards Zero-Shot Point Cloud Registration across Diverse Environments
- Jul. 2021 – Jan. 2023: Absence for mandatory military service

Sejong Science High School Mar. 2016 – Feb. 2019
Specialized High School for Gifted Students in Science and Mathematics Seoul, Republic of Korea

Research Interest

- **Geometric Feature Learning, 3D Reconstruction, 3D Vision for Robotics Applications**

Publications

1. Hyungtae Lim*, **Minkyun Seo***, Luca Carlone, Jaesik Park†, "Toward Zero-Shot Point Cloud Registration Across Diverse Scales, Scenes, and Sensor Setups", *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 48, no. 11, pp. 15234–15251, Nov. 2026. DOI: 10.1109/TPAMI.2026.3722524

2. **Minkyun Seo***, Hyungtae Lim*, Kanghee Lee, Luca Carlone, Jaesik Park†, "BUFFER-X: Towards Zero-Shot Point Cloud Registration in Diverse Scenes", *Int. Conf. on Computer Vision (ICCV), 2025* ***(Highlight)*** [code]

Research Experience

Visual & Geometric Intelligence Lab (PI: Jaesik Park) | SNU Sep. 2025 – Present
Graduate Student Researcher Seoul, Republic of Korea

- Extended BUFFER-X to object-scale registration and heterogeneous sensor configurations (e.g., Ouster OS0-128 to Velodyne VLP-16)
- Proposed BUFFER-X-Lite, an efficient variant with early-exit strategies and fast pose solvers, reducing computation time by 43% while preserving registration accuracy
- Investigating cross-session map matching for multi-session localization and navigation using learned geometric descriptors

Visual & Geometric Intelligence Lab (PI: Jaesik Park) | SNU Dec. 2023 – Aug. 2025
Research Intern Seoul, Republic of Korea

- Curated 11 benchmark datasets spanning indoor/outdoor environments, 6 geographic sites, 12 sensor types, and scales from 3.5m to 240m
- Developed a zero-shot point cloud registration pipeline with adaptive voxel sizing and search radii, enabling robust cross-domain generalization without retraining

AIoT Lab (PI: Hyung-Sin Kim) | SNU Jun. 2023 – Sep. 2023
Research Intern Seoul, Republic of Korea

- Built ES-Studio, a controlled real-world studio for data collection with adjustable environmental lighting and camera sensor parameters
- Conducted preliminary experiments on covariate shifts across environmental and sensor domains

Academic Service

Reviewer: TPAMI 2026, NeurIPS 2026

Awards and Scholarships

ICCV 2025 Highlight Paper Oct. 2025

AI SeoulTech Research Fellowship | Seoul Future Foundation Jun. 2026 – Present

BK21 Outstanding Research Talent Fellowship | SNU Mar. 2026 – Aug. 2026

Graduate Research Fellowship | DB Group Sep. 2025 – Present

National Excellence Scholarship | Korea Student Aid Foundation Mar. 2021 – Jun. 2024

- Full tuition awarded to the top-performing students in science and engineering across the nation

Merit-based Scholarship | SNU Sep. 2019 – Feb. 2021, Jul. 2024 – Dec. 2024

Skills

Language: Korean (Native), English (Fluent)
Programming Languages: Python, C/C++, Java, Javascript, RISC-V
Frameworks/Misc: Pytorch, Pytorch3D, Open3D, CUDA, Diffusers, LangChain, Git, Verilog
3D Tools: Rhinoceros 3D, AutoCAD, Solidworks, Meshlab, MeshRoom, Blender